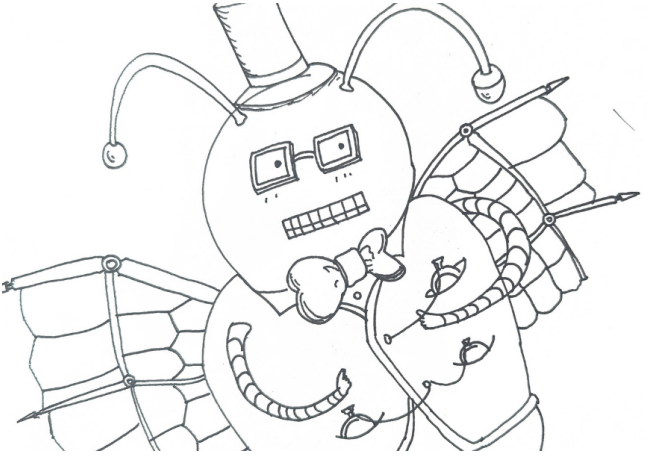
Gerald and the Many Unsettling Details

A Story About the Photoelectric Effect

The world is terrifying, Gerald, the tinker bug, thought, trudging around the castle, staring up at its walls and squinting at some very worrying cracks. His antennae twitched as, in his mind, the cracks widened, shooting through the castle like lightning as the whole thing crumbled.

Gerald yelped and blinked, dropping his tiny clipboard.

The castle was fine. It had not split apart. The tiny cracks were still just tiny cracks.



In fact, the building hummed happily, little electrical currents zipping here and there within its walls. Little loops of charges marched around in their comforting circles.

One of the many advantages of being a tinker bug—which is to say, a bug the size of a small-ish person with large wings, antennae that ended in little glowing balls of light, and a dapper sweater vest, top hat, and monocle—was the ability to hear electrical currents. One of the disadvantages was a tendency to imagine how things could go wrong.

Gerald picked up his clipboard, dusted it off, and made a neat check mark. The crack in the east wall was still only eleven and a half centimeters long.

From there, he moved on to inspect the lemon trees, which supplied the lemon batteries which powered the castle where he and his human friend, Charge, were the sole two apprentices to the mysterious owner of the castle: The Sorceress of Circuits.

He frowned and chewed his tiny pencil. There were only a few ripe lemons today. It had been unseasonably cold and wet recently, and the lemons weren't doing well. The trees looked fragile.

What would they do if they ran out of lemons? The castle would no longer hum with electricity. The lights would go out. The security system would shut down.

Gerald worried, not for the first time, that lemons might not be the most reliable source of power for an entire castle.

He peered more closely at his notes. Hmmm . . . that was odd . . .

The barometric pressure had dropped twelve millibars from the previous day, and the temperature had plunged a startling six degrees. And . . . Gerald sniffed. The air had a sharp metallic tang. He blinked up at the flag flapping in the breeze overhead. It depicted a glowing orange cat with a plus sign on its fur and a tiny helmet with a fish skeleton. The flag

was pointing due south. The wind was coming from the north. Behind the bright orange flag, the clouds thickened into a greenish gray.

Gerald tugged his antenna. This could only mean one thing.

A storm was coming. A bad one.

He made a quick double-check of the lemon trees, grabbed the last few that were ripe enough to add to his emergency stockpile, and then hurried back to the castle.

"OH HEY, GERALD!" his co-apprentice, Charge, said brightly when he bustled into the workshop.

Charge was a human, and she wore a pair of overalls and a wide-brimmed hat with a tiny piece of metal shaped like lightning hanging off the end. She was leaning over a table strewn with wires and lemons. Next to her stood their mentor: The Sorceress of Circuits.

Gerald often wondered how the Sorceress had come to live at this castle. He'd first met her at an inventor's market in a nearby town. He'd been a shop assistant then, and they'd got to talking about the benefits and drawbacks of different wire thicknesses and eventually she'd offered to make him her apprentice.

The Sorceress was a very famous inventor, and Gerald had accepted immediately.

He'd met Charge his first day at the castle. She was clever and quick and impulsive, and he liked her very much although he often found her startling and occasionally intimidating.

Other than the cations, they were the only inhabitants of the castle.

A Note from the Librarian: What are cations? In the real world, they're extremely tiny, positively charged particles. You've never seen one, because they're way too tiny to see, but you've felt their effects:

- *Any time you take clothes out of the dryer, and they're stuck together, and they zap when you pull them apart*
- *Any time your hair gets all staticky when you pull a sweater on or rub a balloon on your head*
- *Any time you scuff your shoes on the floor and then touch a doorknob and get zapped*

Each of those times, you're seeing the results of collections of little charges. There are two types: positive and negative. Like charges repel, and opposites attract. So, two negatives will repel. Two positives will also repel. But a positive and a negative will attract.

To be more specific and science-y, all matter is made up of extremely tiny particles called atoms. Those atoms are made up of

even tinier particles called protons, neutrons, and electrons. And those atoms are made up of a central nucleus which has positively charged particles (protons) and non-charged particles (neutrons) in it. Around that nucleus are the even tinier negatively charged particles (electrons).

Usually, atoms have the same number of positive charges as negative charges. Hydrogen atoms, for example, have one proton in the middle with one electron around it. Oxygen atoms have eight protons all clustered together with some neutrons like a clump of frog eggs in the middle, and eight electrons around them.

Because they have the same number of electrons and protons, the charges are balanced, and overall the atoms are neutral (not charged).

But atoms can lose their electrons. Chlorine, for example, will often get an extra electron attached to it, so it becomes negative. We call that a Cl^- ion. The Cl stands for chlorine, and the – means it has a -1 charge. We call these negatively charged ions 'anions'.

Sodium, which is in the salt you put on your food, often loses an electron, so it becomes positively charged. You might see it written as Na^+. The Na stands for sodium. This positively charged ion we call a 'cation'.

Here in the magical world of the sorceress, she has magically turned some positively charged ions into actual cats. They're not actually cats in our world, sadly.

"HOW DID YOUR INSPECTIONS go?" the Sorceress asked. She had pointed ears and lilac eyes, and she smiled warmly up at him. Spools of wire and a few lemons were scattered across the desk in front of them.

"There's a—a storm coming," Gerald said through his tight throat. "It looks like it'll be bad. I've never seen readings this—" He forgot he was speaking as he noticed three cations at the far end of the workshop. Normally, cations were positively charged ions, atom-sized and invisibly small. The Sorceress loved charges so much that she had turned them into glowing orange pet cats. They were sitting on a window seat, staring out the mullioned windows, their tails flicking back and forth and their ears perked. That was odd, Gerald thought.

"Oh!" the Sorceress said, straightening and laying down her screwdriver. "A windstorm?"

"Or a snowstorm?" Charge asked.

Gerald pulled his attention back to them with difficulty. "Oh, yes, well, both unfortunately. High chance of precipitation. Gusts over a hundred kilometers per hour, I'd imagine, with sustained speeds of at least sixty. The

windows might get broken by flying debris, and the lemon trees are very vulnerable."

"Right," the Sorceress said. "Thank you, Gerald. I'll check on our emergency supplies."

"And maybe we should warn our neighbors," Charge added.

Gerald couldn't quite hear what they were saying over the buzzing in his ears. This was not the comfortable sound of electrical currents. This was his mind buzzing with fear. How did Charge and the Sorceress do it? They seemed so calm, coming up with solutions.

Think, Gerald thought. We need to do something.

Gerald shuffled his papers, and his tiny clipboard clattered as he set it on the table with shaking hands. If the lemon trees are damaged, how long will our battery supplies last? If it gets too cold, the exterior pipes will freeze and maybe burst. Do we have backup pipes? He couldn't remember.

His ankle tingled and he looked down to see a cation weaving around his ankles.

It arched its back, and he gave it a friendly pat. When he looked up, both Charge and the Sorceress had gone.

Gerald adjusted his tiny top hat, tugged on his vest a few times, checked his pocket watch,

tugged on his vest once more, and then hurried after them. He would find some way to help.

HE FOUND THE SORCERESS in the main entryway of the castle, setting up piles of blankets. A herd of cations trailed behind her, sniffing at the blankets and hopping on and off her shoulders.

"Ah, Gerald, perfect. Will you please help me set these up?"

"Of course, of course," Gerald said, glad to have something to do. The wind howled eerily around the castle walls, and the skies darkened. Gerald shivered.

He was just setting up the last of the piles of blankets when the entry door banged open and Charge appeared, silhouetted against the glowing gray-green sky. A flurry of snowflakes swept in on a gust of frozen air. Gerald wrapped his arms around his stomach and shivered.

"Come on, inside everyone, inside!" Charge said brightly, waving in fifteen deer, two rabbits, twelve ducks, a fox, and one very sleepy-looking bear.

A gust of wind blasted through the open door, sending the chandeliers swinging, and everyone gasped as the power flickered.

Charge threw her weight against the door, slamming it shut. Gerald stared at the delicate flakes of snow as they melted into quaking droplets.

"Thank you for having us," one of the ducks said, approaching Gerald. It shook some of the snow off its back. "Very odd weather we're having."

"Indeed," Gerald said absently. "Very odd." His mind was spinning through all the contingencies. All the things that might happen. They had enough lemons to last a few days. They had enough glass to repair at least nine broken windows. But that crack. That crack really worried him. Was the castle strong enough? He readjusted his top hat. Silliness, he said to himself. Pure silliness. This castle has stood strong for centuries! It must have seen worse storms than this!

The Sorceress laid a hand on his shoulder. "Don't worry, Gerald," she said. He glanced up at her. She was smiling warmly down at him.

Something crashed into the castle wall with a loud bang, and she winced. Her smile quickly returned, and she gave his shoulder a reassuring pat. "This is a strong old castle," she said.

Gerald clenched his trembling fingers and fluttered his wings. The sky outside was com-

pletely dark now, and the howl of the wind rose and fell and rose again.

Something banged off a window in the dark and Gerald flinched.

In the middle of the room stood Charge, directly under the chandelier. Her fists were braced against her hips, and her hat tipped back as she stared at the windows. She looked ready for anything. Ready to spring into action at a moment's notice. The ducks clustered around her as if they sensed that she would be one to have useful ideas in a crisis.

I wish I could be more like Charge, Gerald thought. She doesn't lock up when she gets scared. She leaps into action. How does she have so many ideas? My mind just goes blank. Isn't she afraid at all?

A great snore issued from behind one of the pillars. A giant bear paw stuck out from under a pile of what looked like most of the blankets. The soft mountain shook with the rumbling snores. The bear was clearly not afraid at all, either.

The Sorceress squeezed Gerald's shoulder gently. "I'm so glad you noticed the storm was coming," she said. "You gave us time to prepare."

The lights at the ends of Gerald's antennae glowed a little brighter at this. Even the wind outside seemed to die down. His shoulders relaxed, lowering from his ears. Maybe the storm was slowing down. Maybe it wouldn't be as bad as he'd predicted.

Something slammed into one of the windows, shattering it. Broken glass rained onto the stone floor, and freezing air and snow blasted into the room. The ducks quacked in alarm, flapping their wings and gliding for cover behind the bear.

The Sorceress's hand left his shoulder as she ran for the broken window. Charge was already there with a ladder and a large board. The Sorceress produced a hammer from somewhere in the folds of her dress, and in a flash, Charge was up the ladder, hammering the board into place.

Gerald's tiny feet remained rooted to the stone floor beneath him. Again, his mind was buzzing. How could they move so quickly?

The cations didn't seem to mind the cold. They hopped to and fro in the snow, batting bits of it at each other even as it melted.

Gerald fluttered his eyelids and sucked in a deep breath.

I have to do something, he thought.

How come every time something startling or scary happened, all he could do was stand there and watch it? It was lucky that Charge and the Sorceress didn't have the same problem. If they weren't here, how long would I have stared at the broken window? Gerald wondered. It would have gotten very cold in the castle by the time he'd been able to think of what to do. And that would have cost them valuable energy. Heating the castle on only lemon battery power was extremely lemon-intensive. They should probably have some sort of backup power source, Gerald thought again. It took hundreds and hundreds of lemons to warm the castle up. And lots and lots of copper and zinc bits for the electrodes of the lemon batteries.

> *A Note from the Librarian: Can you really make a battery out of a lemon? Yes, you can! It doesn't provide very much power, though.*

Thinking of the batteries reminded him of the creepy basement where they kept their metal stores, and he shuddered. The basement was absolutely his least favorite part of the cas-

tle. He never went there if he could help it. Never even did any checks. If he needed to know how much zinc or aluminum or copper or iron they had left, he always asked Charge to go check.

And now he wondered. Would they have enough zinc and copper for their batteries? He knew they had approximately fifteen days of lemons for lemon batteries stored. Although if windows kept breaking and they had to keep warming the castle back up again that might only last them . . . he did some quick mental calculations about the thermal conductivity of the castle walls and the energy transfer efficiency of their heating coils. Seven or eight days. When was the last time he'd asked Charge to check on the zinc and copper stores?

His wings went still, and his stomach twisted. It had been at least a month. He tried to think of the basement as little as possible.

And now look! What if, because of his fears, they were close to running out of the metals they'd need to keep their batteries running through this bad weather?

A deer stared, wide-eyed, at him from behind a nearby pillar. Two fuzzy baby ducks peeked out from behind her legs. What if they ran out of power and it was still cold out and

they couldn't keep the castle warm? It wasn't usually so cold this time of year. The baby ducks weren't prepared for it. They needed to be kept warm, or they might not make it. Gerald clenched his tiny fists. He wouldn't let anything happen to the baby ducks.

Another window shattered, and another, colder gust of air whirled into the hall. Charge and the Sorceress were there in a flash, their tools flying as they boarded it up. But the board wasn't sealed well against the window. The drafts were like cold, invisible fingers tugging at Gerald's ankles.

I'm not helping here at all, Gerald thought. His antennae dropped, his wings hung slack below his shoulders. Would he freeze here in place forever? Become a little tinker bug statue?

Maybe they would put a little sign on him. This is a statue of a tinker bug who thought and thought but was too scared and wasn't able to help his friends when they needed it. No, they wouldn't do that. That would be mean, and Charge and the Sorceress were both too kind to ever say anything like that to him. Which made him want to help even more!

I could go check our metal stores, he thought. His stomach constricted, and his

mind spun in terror, imagining the dark, creepy basement with all its strange sounds and shadows. He couldn't go into the basement. He couldn't.

It was the worst possible thing he could imagine.

But.

Maybe if I go into the basement, it'll be so scary that nothing else will seem scary by comparison. Maybe if I go do the scariest possible thing, I won't be afraid anymore. And then I won't freeze up anymore. I'll be able to be like Charge and the Sorceress, reacting quickly, having good ideas even when things are going terribly. Even when it's dark and stormy, and invisible things are smashing windows in the cold.

Gerald hated this thought.

He hated that it seemed like a good idea.

GERALD TOOK ONE LAST look at Charge and the Sorceress. They were busy hammering boards into place, and neither of them looked up as he crept off down the hallway that led to the creepy basement.

All too quickly, he was standing at the top of the stairs that led to the basement. Gerald sniffed the faint, musty draft emanating from

the darkness. I could bring some cations for light, he thought. He straightened his vest. No. That would make it less scary.

His tiny, polished shoes clicked on the stone steps as he descended into the dark.

The air became colder and damper as he went. His whole body was tight, alert, listening for the slightest sound.

Halfway down the stairs, a droplet of water dripped somewhere in the darkness. The sound echoed through the stone passage. Instantly, Gerald imagined an open mouth, full of sharp teeth, the face of some creature shrouded in shadow. Saliva dripped from between the teeth into a puddle below.

Gerald squeezed his eyes tight and then opened them again. There was nothing to see. It's just condensation collecting on the walls and ceiling and then dripping, he told himself firmly. There's not some creature sitting down here drooling constantly. That's just illogical. It would become dehydrated.

Unless it's drooling because it sees me, his mind insisted.

It took Gerald several seconds to get the courage to descend again into the dark.

After what felt like an eternity, he reached the stone floor. He could feel the space stretch-

ing out around him. Something about the way every sound echoed gave the sense that he was in a very large room. The light from his antennae made a dim halo that was quickly swallowed by the vast, dark space.

I should go back upstairs. They might need me.

He turned and darted a few feet back up the steps but then stopped. He took a long, deep breath of the musty air. He tasted the tang of metal. All the metals that were so important to his work. All the incredible things the Sorceress had invented and would invent in the future. This was such an important part of what they did here. He didn't want to run from it.

With a quick adjustment of his top hat, he turned and marched back down the steps and straight into the towering darkness.

It closed in around him. The looming shapes of hulking sheets of metal rose up around him like teeth but he strode past them, his fists clenched and his wings trembling.

He stopped, breathing heavily, in what he approximated was the center of the room. The air was strangely warm, almost hot, and he made a mental note to check the heat settings down here. Something tipped over nearby, clat-

tering to the ground, and Gerald shrieked. His tiny calves clenched, preparing to sprint for the exit. He pictured hands reaching out to grab him, a mouth widening, preparing to gobble him up.

But he didn't move.

He stood craning his neck, his head tilted to the side, listening. The sounds of the electric current were so distant now, muffled by layers of rock, and here in the basement, aside from the occasional drip—which was definitely condensation and not saliva—it was very quiet.

> *A Note from the Librarian: Electric current itself doesn't actually make any sound. (Although, sometimes you can hear a buzzing if the wires themselves are vibrating.) There are some animals, though, that can sense electric and magnetic fields. An electric current creates a magnetic field around it, and some animals like sharks can sense it. Similarly, bees can detect electric charge with specialized hairs on their body. In this story, Gerald is a tinker bug, which doesn't exist in the real world, but in this story, he too can sense electricity, charges, and the magnetic fields surrounding electric currents. The biological term for this is electroreception, but since we humans can't do it we don't have a good word to describe what this sense is like. So, in this story I'm*

> *describing Gerald's sensation of current as 'hearing' and sometimes 'seeing', but he's not really hearing or seeing the current, he's using electroreception to pick up the magnetic field surrounding the currents.*

AS HIS EYES ADJUSTED to the darkness and his breathing slowed, he began to look around him. The light from his antennae reflected off the stacks of metal. The Sorceress was always heading off on foraging expeditions, and many traveling tradespeople would collect interesting bits of metal for her, stopping by to offer her some in trade. There were stacks of bright silver aluminum, dull oxidized copper, partially rusted iron. There might not be monsters down here, but the metal itself was alive in a way. Only the bits of gold sat unchanging. The rest of it reacted with air, developing a dull coating, or rusting away into red powder. He loved especially the turquoise color of the oxidized copper.

> *A Note from the Librarian: These are all real metals, and real properties that they have in the real world.*

HE'D FORGOTTEN HOW MUCH beauty there was down here. Forgetting also that he was on a mission to be as terrified as possible, he turned up the brightness of the glowing ends of his antennae, admiring the way the light reflected in bright bands off the many-colored metals.

The yellow light from his antennae poured throughout the room, and the sharp ends of metals cast long shadows down the aisles.

A few rows farther up there sat a lovely huge pile of oxidized copper ore. He folded his hands across his vest and puttered over, peering down at it.

Zzzap!

Gerald leapt into the air and stumbled backwards, spinning his head left and right and fluttering his wings. Where had that come from?

Taking a deep breath and holding it, he stood frozen, listening. But there was nothing.

At last, curiosity got the better of him, and he stepped back towards the copper.

This time, because he was paying attention, he heard the low buzz right away. It wasn't coming from the copper, though.

He glanced sideways down the aisle.

There, at the far end, barely illuminated by the light from his antennae, was a strange golden liquid, submerged in oil, and Gerald knew it immediately. Cesium. Cesium was a very soft metal at room temperature and melted into a liquid at twenty-eight degrees Celsius, around eighty-three degrees Fahrenheit. If the cesium was in its liquid form, it must be quite hot down here indeed.

It had to be kept submerged in mineral oil because if it came into contact with air, it would oxidize very rapidly, and if it came into contact with water, it wouldn't just rust or develop pretty blue patterns. It would explode, fizzing and releasing hydrogen gas, which would ignite into a ball of fire. Such a terrifying, dangerous metal, and Gerald did not want to get anywhere near it.

However, the buzzing sound was coming from the cesium.

> *A Note from the Librarian: Cesium is a real metal, too! And everything that cesium does in this story is something the real cesium in the real world does. It's just that Gerald is able to perceive it directly, which we wouldn't be able to without scientific instruments.*

AND IT WAS THAT buzzing sound, only heard by tinker bugs, that meant electrical current.

It was not a good sign that the dangerous, explosive metal that had to be kept in a special container was somehow giving off an electrical current. This could be an even worse problem than the storm.

Examining it carefully from his distant vantage point, though, he couldn't see anything wrong with it.

Well, doing scary things is what I'm here for, Gerald thought. Although, if the metal showed any signs of exploding, he would stay far away. But the glass vat looked solid. No cracks or leaks.

He moved down the aisle, listening carefully, his eyes locked on the cesium.

When he was a few meters away, he stopped. On the stone ledge next to the vat with the cesium were two more vats, each with large chunks of silvery metal in them. Potassium and sodium. Both were explosive and would oxidize in air and explode violently in water, although neither was as bad as cesium.

> *A Note from the Librarian: This is all real, too! Potassium and sodium are both real metals in the real world. You can find them on the periodic table of the elements. They're never found in their pure form in nature, though, because they react so quickly, oxidizing in air and exploding violently and releasing hydrogen gas if exposed to water.*

PURE SODIUM, GERALD THOUGHT, staring at it. He marveled for a moment that this violently explosive metal could be made into simple table salt if you just added some chlorine to it. NaCl. Na for sodium, and Cl for chlorine. He imagined what would happen if he accidentally salted his soup with actual sodium, and not sodium chloride. His soup would explode. Thank goodness for chlorine, he thought, not for the first time.

The faint electrical buzzing brought him back from his fascinated thoughts. It was louder now. Definitely coming from the cesium. Little electrons pinged off it, zipping around. But he couldn't see anything out of the ordinary that might be causing this.

Wanting to look more closely, he turned up the brightness of his antennae, and the yellow light shone brightly on the metals.

The zipping and pinging intensified, like a great fizzing of electrons popping off.

> *A Note from the Librarian: Electrons are way too tiny to see. They're extremely tiny, negatively charged particles that are part of atoms. In this story, because Gerald can sense charges, he can sense the presence of electrons. In the real world, we might notice some static electricity or a spark if there were enough electrons moving.*

GERALD QUICKLY DIMMED HIS antennae. The zipping slowed.

Is it my light that's causing this? he wondered, fascinated. He'd never heard of light being able to create current. How marvelous. How utterly—a bright orange ball of glowing fur appeared at his side, mewling. The cation hopped curiously up onto the ledge and batted its paw at the vat of cesium, as if it were a barrel full of fish.

"No, no, not for you, sorry," Gerald said, picking up the cation and setting it down. They were always attracted to negative charges. The cations were curious about current, too, and static electricity as well. All sorts of negative charges, stationary (static) or moving (current).

The curiosity of the cations was yet another sign that the cesium was somehow giving off electrons—tiny negatively charged particles—when he shined his light on it. The cation meowed, winding around Gerald's legs and begging to be allowed back onto the ledge. Gerald patted it distractedly, still staring at the cesium.

This was something unexpected. And that meant he might learn something. And when there was something to learn, there might be something to invent.

Even though this seemed like such a tiny detail, he knew it was a doorway to the unknown. To magic. To amazing inventions he couldn't even imagine or predict.

And that meant it was time to experiment.

Gerald by now had forgotten all about the storm and about being afraid. He had a problem to tinker with. And tinkering was what he did best.

The first question was, why only the cesium? Would other metals do this, too?

Maybe he needed to make his lights brighter.

Cautiously, he turned up the brightness again. The current grew larger and louder. More and more electrons poured off the

metal. The cation yowled, and two more appeared at the base of the stairs, streaked across the basement, and the three of them hopped up onto the ledge, their tails flicking back and forth as they pressed their paws against the glass.

> *A Note from the Librarian: Because electrons are so incredibly minuscule, there are a LOT of electrons. Everything you can see and touch has trillions and trillions of electrons in it. So, even though electrons are coming off the cesium, it's tiny compared to how many electrons are actually in the metal.*

BUT NONE OF THE other metals reacted.

Gerald quickly dimmed the light, and the cations looked from each other to him in disappointment. One of them batted its paw against the glass experimentally, but the jar was bolted solidly to the ledge.

Gerald double-checked that all the jars were securely fastened and then continued.

What else could he change? Maybe his lights weren't bright enough. He was afraid, though, of what the cesium might do if the current coming from it got any higher.

Brightness wasn't the only quality light had, though. It also came in a variety of colors.

Being a tinker bug, like some other bugs, Gerald could see all the colors human beings could see, plus a great many more. Like ultraviolet, which was just past purple, just out of the range people could see. And infrared, which was on the other, less energetic side of the electromagnetic spectrum.

Light was something both he and the Sorceress were fascinated by. The Sorceress loved electricity, and magnetism, and light was an electromagnetic wave. An oscillating field that was first an electric field and then a magnetic field, and then an electric field again. Unlike sound, it didn't need air—or anything—to travel through. It could just zip along in a straight line forever until it ran into something.

And, just like all waves, it came in different types. Different energies. Which people saw as colors.

He'd explained it to Charge once by borrowing her jump rope. He'd placed the rope on the ground and asked her to hold one end still, and then he'd held the other end. First, he wiggled the rope back and forth across the ground slowly. The rope had made a wiggling wave pattern which moved along the rope towards

Charge. Then he'd wiggled the rope faster and faster. The waves had gotten closer together, smaller and tighter.

"The faster I move my hand," he'd said to Charge, "the more energy the waves have."

"They do look more energetic," she'd answered, staring at the wave pattern.

"We call this the frequency of the wave," Gerald had explained. "Which is really just a fancy way of saying, how many waves there are every second."

"Oh, so if you wiggle your hand over and back once in one second, that's a frequency of one?"

"Precisely," Gerald had said, moving the rope slowly, so that his hand went over and back once in a second. The rope made big, lazy waves. "And if I move my hand quickly, say ten times in a second, that's a frequency of ten." He shook his hand over and back, trying to do five wiggles in a single second. "We measure frequency in 'Hertz' but it really just means 'per second'."

"Gotta sound fancy, I guess," Charge had said.

Gerald had agreed, although he felt like sometimes, even though it was good to have precise words for things, it sometimes made

things sound more complicated than they really were.

One of the cations hopped up onto the top of the cesium vat and began sniffing around the lid, which was locked in place, but still it was making Gerald nervous. He plucked the cation off and set it on the ground, where it yowled at him.

"Perhaps if I change the color," he thought aloud. One of the cations gave him a curious look.

Light waves had much, much higher frequencies than string waves did, but the principle was the same.

"What color should I try . . ." he wondered, tapping his chin with one finger. The yellow light he'd been using had a frequency of over 517 trillion waves per second. Which was mind-boggling. He couldn't even imagine something wiggling back and forth that many times each second. Technically, because it was light, it was those electric and magnetic fields pulsing. But still.

"Shall we try green light?" Gerald asked the cations. They stared up at him, but when he didn't produce any anions or fish, they turned back to stare at the vats.

Green light was slightly more energetic than yellow light. Depending on the shade, it might have a slightly higher frequency of around 545 trillion waves per second.

Ever so carefully, he increased the frequency of the light coming from his antennae from yellow, through a pale lime green, to a full bright green. He kept the brightness the same, changing only the color.

The cesium continued to zip and ping. It sounded like the same amount of current, the same number of electrons. But they shot out faster and faster. As he increased the frequency of the light, he seemed to be increasing the energy of the electrons released, too.

And when he hit green, something very unexpected happened.

The sodium began to buzz with current, too.

The cations yelped, their heads flicking back and forth between the two vats as if they were watching a tennis match.

Gerald stood, open-mouthed, staring at the vats. Both metals were now sending little electrons flying off their surfaces. The ones from the cesium were flying considerably faster than the ones from the sodium, but they were both shooting off electrons.

"How in the world . . ." Gerald muttered.

Somewhere far overhead, there was a dull thud and a bang, but Gerald barely noticed. He was too deeply immersed in the mystery at hand.

To confirm his earlier results, he tried turning up the brightness of his lights again. Yet again, the electrons were still travelling at the same speed, but more and more of them were knocked off of both metals this time.

"But, this is so strange," Gerald said to the cations, who were completely ignoring him, hopping from side to side, batting at the vats and then leaping backwards and away. "Light is a wave. Waves of greater intensity are usually a greater energy, too. So. I would have thought that making the light brighter would make the electrons go faster. But it doesn't. The energy the electrons have depends on the color. And also whether or not they are released depends on the color, too." He scratched his head, knocking his top hat askew.

Curiously, he turned the color from light green to a slightly higher-energy blue-green. The potassium began to emit current, too!

The electrons from the sodium were now speeding out even faster, and the electrons

from the cesium were zipping away even faster than that.

"This is very strange," Gerald said to himself. "This is not what I expected at all. That must mean something about my understanding of light is not quite right, too. But what?"

> *A Note from the Librarian: In order for light to knock electrons off a metal, the light has to have enough energy for that particular metal. We call this energy requirement the 'work function' of the metal. In reality, sodium and potassium have very similar, overlapping work functions. But it also can vary from surface to surface depending on contamination or the orientation of surface crystals. Georgia State University's list of work functions on HyperPhysics lists sodium's work function as 2.28 electron-volts (eV, a unit of energy) or 2.36 eV and potassium's work function as 2.3 eV or 2.29 eV depending on the type. So, they might have started at the same time, or in either order, depending on the specifics of these particular samples that they have in their basement.*

ABSENTLY, WANTING MORE DATA, he turned the color of his lights from blue-green up to blue. Then to indigo, and finally to violet. The electrons from all three metals sped

up, but no more were released unless he made his lights brighter.

"It's as if . . . as if there are little balls of energy flying at the metal," he said. "And that, in order to knock off the electrons, the balls need a certain amount of energy. If they have enough energy to knock them off, then they will, and any extra energy they have makes the electrons go faster. But in order to knock off more, I need to turn up the brightness. I thought light was just a wave, but . . . it's almost like it's made of little particles, little packets of energy, too. How very odd." Gerald stood in the dark basement, which had until very recently been the most terrifying place he could imagine, listening to the little electric currents, which only he could hear, and wondering what this meant for other things he thought were true.

Spending time in the most terrifying place he could imagine was turning out even better than he'd hoped! But for an entirely unexpected reason! He'd had no idea there would be a scientific puzzle down here, or he would have come down sooner.

Another distant thud brought him back to reality, and he remembered the storm raging overhead. The winds were so high, they must

be damaging the lemon trees. He hoped they made it, because if they didn't, they were going to need some other source of power.

Could his discovery help them somehow? Electricity was just a bunch of electrons flowing through wires. And here he had noticed that he could knock electrons off of metal just by shining light onto them. And there was—usually—lots of light available outside every day. Could they just put metal out in sunlight and make current that way?

He sensed a tinker-challenge. Because so often, he would get a little idea, or learn something new, and it would be like an amazing flash of brilliance. But then, to actually make the idea into something workable, it would take a vast amount of trial and error. Of tinkering. He rubbed his hands together. There was nothing he liked better than tinkering around with something, trying hundreds of variations to get it right.

Inventing light bulbs had been the same way. He and Charge and the Sorceress had tried hundreds and hundreds of combinations of little metal filaments, seeing which combinations worked the best. They would find something that sort of worked, but was way too hard to make, or too expensive, or required

materials that were too hard to get. So they'd try something else, making it just a bit better and then just a bit better again. Often it took them years and years and years of tinkering.

> *A Note from the Librarian: This is similar to what happened in the real world. Thomas Edison didn't actually invent the light bulb. Light bulbs had existed for decades already, but they were too expensive, and they didn't last long. He was able to make a version that lasted long enough and could be mass-produced cheaply enough that people could actually buy and use them.*

THAT WAS WHY THERE were so many half-finished projects around the castle. Because there were so many things they were playing around with, seeing if they could make it just a tiny bit better. Sometimes he got overwhelmed with all the different projects, but as soon as he got absorbed in one in particular he would forget all about that. He loved the process of slowly playing around with something, trying to make it better.

Gerald sensed this would be the same. There would be a lot of details to work out.

For instance, he really, really did not want to have to work with these exploding metals.

He tugged on one of his antennae thoughtfully. He'd tried the colors that humans could see, but there were so many more. Gently, he turned his lights to a bit higher of a frequency. Past what humans could see, all the way into the ultraviolet range. A little further. A little further still.

Behind him, the aluminum began to sing.

His eyes widened.

Curiously, he pushed just a bit further.

The copper joined the chorus.

Gerald did a little dance in his tiny, polished shoes as the cations ran from metal to metal, hopping this way and that, trying to catch the invisible, impossibly tiny electrons. Copper would be much easier to work with than the exploding metals.

The fact that those electrons went flying off everywhere might be a problem, though. He'd need a way to catch them and direct them if he wanted to create current. He couldn't just have them flying everywhere.

He knew there was still a vast amount of trial and error in front of him, but he suspected he had found something he could use to generate current someday. They would no longer have to worry if the lemon trees were damaged!

There was another bang overhead, and he lowered his lights back to the visible spectrum and bustled towards the stairs.

He wasn't fleeing in terror this time. And he didn't even see the little creature that waved and smiled at him from the corner, the drool dripping from its teeth as he ran past.

"I've got it!" Gerald shouted, fluttering his wings as he burst into the great entry hall of the castle.

To his surprise, bright sunshine streamed in through the windows. Only three were broken, haphazardly boarded over.

The hall was empty.

A shout echoed in through the broken windows, and Gerald fluttered across to the heavy oak doors, tugging one open and squinting as the sunlight warmed his face.

The world sparkled. The sky overhead was empty of clouds. The storm had blown them all away, and the bright blue expanse stretched overhead, the morning light glittering off the newly fallen snow.

"Gerald!" Charge shouted, running over to him and grinning. She flopped onto her back beside him, waving her arms and legs and then

leaping back up again, grinning down at the snow angel she'd made. "Where were you?"

"Oh, well," Gerald said, his hand going to his tiny bow tie and adjusting it nervously. The rabbits were hopping around in the snow, and the heavy tracks of the bear angled off into the woods. "I was, er, I was down in the basement."

Charge's eyes widened. "The creepy metal basement? But you hate it down there!"

The Sorceress tramped up, her cheeks bright from the cold, her breath sending up clouds of steam as she smiled at them. "What's this, Gerald? You were down in the basement? I'm sorry, you must have been more stressed than I realized with the storm last night, huh?"

Gerald's face heated, and a small lump formed in his throat. He loosened his bow tie, but it didn't help. "Oh, well, hrm. I suppose. I . . . I went to check on our metal supplies." He stared down at the bright snow, so friendly and safe seeming now in the morning light. Time sure had passed quickly in the creepy metal basement.

The weight of their curious stares pressed down on him. "Well, to be honest, no. I . . . I felt terribly that I was not any use last night. You both reacted so quickly to those broken

windows and I . . . I simply stood there." His throat tightened, and he couldn't bring himself to look at them.

The Sorceress placed a hand on his shoulder, and Charge gave his head a quick hug and then released him.

"Aw, Gerald, you're so hard on yourself," Charge said. "Sorry, did I hurt your head?"

"No, no, it's fine," Gerald said, taking the fallen top hat that Charge held out to him.

"But why the basement?" the Sorceress asked.

"Oh, well, I considered it, and I thought perhaps if I went someplace very, very scary, nothing else would seem quite so scary by comparison."

Their shadows on the bright snow nodded, and he looked up at them. Their faces were sympathetic.

"How was it?" Charge asked.

"Oh, horrible," Gerald said with a shudder. "But also, incredible!" He told them of his discovery.

"Metals give off electrons when you shine light on them?" the Sorceress gasped, her eyes glowing almost brighter than the snow.

"That's incredible," Charge said.

"Indeed," Gerald said. "I think perhaps we could use this to generate current, eventually. After some experimentation, of course."

"Well, that's extremely lucky timing," the Sorceress said, "because the lemon grove is . . . a bit chaotic at the moment. I think they'll pull through, but it might be a while before we have lemons."

"Oh, er, well, I think it might be some time before I have a working prototype," Gerald said.

The sound of runners gliding over snow and the tinkling of bells echoed through the bright morning. They all turned to watch a brightly painted sled emerge from the trees. It was piloted by a thin and very cranky looking elderly man in a tall, narrow top hat, and pulled by three shiny robots. The sled was piled high with potatoes.

"Professor Watt!" the Sorceress exclaimed as the sled pulled up a few feet from her. "What are you doing here?"

He swept off his hat and laid it on the seat next to him. "Just checking in on my neighbors," he said. "Heard the storm last night and I thought your trees might have had a bad time of it. Thought I'd bring you some backup

potatoes. In case you needed some temporary power."

"Watt, that is so thoughtful of you, thank you," the Sorceress said. "Yes, we were just thinking we might need something to tide us over until Gerald can get his new invention working again."

Watt leaned forward and dripped some oil into the shoulder joint of one of the robots. "New invention, eh?"

Gerald wasn't sure he trusted the rival inventor, despite his general friendliness, but he didn't say anything as the Sorceress explained what Gerald had discovered.

Professor Watt scratched his forehead thoughtfully and stared up into the bright sky. "Well, that is quite fascinating. That does seem useful."

It's only an idea, Gerald thought. Ideas are useful, but it's the long tinkering process that really makes them great. This thought eased his anxiety somewhat.

Professor Watt declined to stay for tea, and after he'd left, Gerald, Charge, and the Sorceress went back inside to wire up the potatoes he'd kindly brought.

"You know, Gerald," the Sorceress said, "it's very lovely that you've made this new discovery, but I do just need to say that you are very skilled and very valued. And we all freeze up sometimes. It happens to me, too."

"And me," added Charge, sticking a bit of metal into one end of a potato and clipping it to an electrical lead.

"We wouldn't even have known the storm was coming without you," the Sorceress said.

"And you collected all the lemons that were ready and brought them inside before the storm, which was why we had enough power to keep the castle warm in the first place," Charge added.

"Yes, but all those broken windows . . ." Gerald said.

"That's what you have us for," Charge said.

"And, if it happened more frequently, I'm sure you would become just as skilled at reacting quickly," the Sorceress said. "You notice so much that Charge and I miss. You're very perceptive, and you see tiny details, and you're very tenacious at following through on your observations."

Gerald's face heated, and he fumbled with the wires he was holding, suddenly forgetting what he'd been planning to do with them.

"I . . . I see," he said. "Well, thank you."

"And more than that, we like you," Charge said, patting him on the shoulder. "You're nice to have around. You don't have to be different."

Gerald was silent for quite some time, thinking this through. Eventually, the conversation turned to the potato-wiring project, and what they might need to do to help the lemon trees make it through the next storm. And Gerald's mind drifted back to thinking about the metals, and how he might combine them to make an efficient way of turning sunlight into electricity.

For now, he forgot about wishing he were different, wishing he weren't afraid. He was just happy to be there with his friends, and with many interesting electrical mysteries to systematically test and investigate.

The End

A Note from the Librarian

Metals really do emit electrons like this when you shine light on them. It's called the

photoelectric effect, and it's the basis for solar panels and part of quantum mechanics. This phenomenon was observed in various ways by a lot of different scientists throughout the 1800s. One of the things that confused people about this observation was that at that time they thought light was only a wave.

If light was a wave, then they should be able to shine a light at a piece of metal and wait a long time and eventually see an electron knocked off. Because the energy from the wave should just build and build until it wiggled the electron loose. Like wiggling a tooth back and forth until it falls out. But, just as Gerald noticed, this isn't what they saw.

Instead, they saw that if the frequency wasn't high enough, then no matter how bright the light was or how long you shined it at the metal, no electrons were knocked off. It was Albert Einstein who wrote a paper in 1905 and suggested that this was because light also could behave as a particle. J.J. Thomson called electrons 'corpuscles' which is such a great word, and Einstein called the particles of light "light quanta" but later we would call them photons.

Einstein suggested that each photon had a particular energy that was related not to the brightness of the light (its intensity) but to its frequency. You can find its energy by simply multiplying its frequency by Planck's constant, h. (The equation for this is $E = hf$*.)*

Einstein received the Nobel Prize in Physics for this work in 1921, and it was part of the work that laid the foundations for quantum mechanics.

This effect is also related to how we make solar panels! You can actually build a really rudimentary solar panel at home with pieces of copper. In solar panels today, we actually use different materials called semiconductors (like silicon), which helps guide the electrons through the material in a controlled way, rather than letting them fly off and escape. But it's a similar idea in that you shine light on a material, and it dislodges electrons, which creates current.